

Kubernetes Day 9 9 June 2026 K8s Secrets

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
```

```
$ eksctl create cluster --name ekscluster9June --nodegroup-name b17dng --node-type t3.micro --nodes 5 --managed
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
```

```
$ eksctl get cluster
```

NAME	REGION	EKSCTL CREATED
ekscluster9June	us-east-1	True

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
```

```
$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
ip-192-168-19-24.ec2.internal	Ready	<none>	6m51s	v1.34.8-eks-3385e9b
ip-192-168-28-183.ec2.internal	Ready	<none>	6m49s	v1.34.8-eks-3385e9b
ip-192-168-32-252.ec2.internal	Ready	<none>	6m50s	v1.34.8-eks-3385e9b
ip-192-168-50-122.ec2.internal	Ready	<none>	6m50s	v1.34.8-eks-3385e9b
ip-192-168-52-111.ec2.internal	Ready	<none>	6m48s	v1.34.8-eks-3385e9b

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
```

```
$ ll
```

```
total 16
```

```
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 13:21 1_june/
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 11:39 29_May/
drwxr-xr-x 1 Bhavana 197121 0 Jun  3 19:56 2_june/
drwxr-xr-x 1 Bhavana 197121 0 Jun  4 07:48 3_June/
drwxr-xr-x 1 Bhavana 197121 0 Jun 11 16:38 5_june_And_9_June/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
```

```
$ cd 5_june_And_9_June/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
```

```
$ ll
```

```
total 15
```

```
-rw-r--r-- 1 Bhavana 197121 691 Jun  9 19:01 1-pod_rr_rl.yaml
-rw-r--r-- 1 Bhavana 197121 481 Jun  9 23:31 2-postgress-pod.yaml
-rw-r--r-- 1 Bhavana 197121 514 Jun  9 23:29 3-pod-env.yaml
-rw-r--r-- 1 Bhavana 197121 295 Jun 11 16:34 4-fb-configmap.yaml
-rw-r--r-- 1 Bhavana 197121 273 Jun  9 23:49 5-fb-pod.yaml
-rw-r--r-- 1 Bhavana 197121 184 Jun 11 16:35 6-secret.yaml
-rw-r--r-- 1 Bhavana 197121 268 Jun 11 16:32 7-fbpod-withsecret.yaml
-rw-r--r-- 1 Bhavana 197121 259 Jun 11 16:45 8-appconfigmap.yaml
-rw-r--r-- 1 Bhavana 197121 594 Jun 11 16:44 9-backendDeploy.yaml
```

```
#6-secret.yaml
```

```
---
```

```
apiVersion: v1
```

```
kind: Secret
```

```
type: Opaque
```

```
metadata:
```

```
  name: fb-secret
```

```
data:
```

```
  Course: QVdTIERldk9wcw==
```

```
  Batch: QjE3RA==
```

```
  Trainer: TWFKaHVLaxJhbg==
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
$ kubectl apply -f 6-secret.yaml
secret/fb-secret created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
$ kubectl get secret
NAME      TYPE      DATA      AGE
fb-secret Opaque    3          12s
```

```
# 7-fb-pod-with-secret.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pod-with-secret
  labels:
    app: fb-app
spec:
  containers:
  - name: fb-secret-container
    image: nginx
    ports:
    - containerPort: 80
    envFrom:
    - secretRef:
        name: fb-secret
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
$ kubectl apply -f 7-fb-pod-with-secret.yaml
pod/fb-pod-with-secret created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
$ kubectl get all
NAME                                READY    STATUS    RESTARTS    AGE
pod/fb-pod-with-secret             1/1      Running   0           10s
```

```
NAME                                TYPE      CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/kubernetes                  ClusterIP   10.100.0.1    <none>         443/TCP    17m
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)
$ kubectl exec -it fb-pod-with-secret -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=fb-pod-with-secret
NGINX_VERSION=1.31.1
NJS_VERSION=0.9.9
NJS_RELEASE=1~trixie
ACME_VERSION=0.4.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
Batch=B17D
Course=AWS DevOps
Trainer=MadhuKiran
KUBERNETES_PORT_443_TCP=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.100.0.1
KUBERNETES_SERVICE_HOST=10.100.0.1
```

```
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.100.0.1:443
TERM=xterm
HOME=/root
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

```
$ kubectl describe po fb-pod-with-secret
```

```
Name:                fb-pod-with-secret
Namespace:           default
Priority:             0
Service Account:     default
Node:                ip-192-168-50-122.ec2.internal/192.168.50.122
Start Time:          Thu, 11 Jun 2026 16:50:07 +0530
Labels:              app=fb-app
Annotations:         <none>
Status:              Running
IP:                  192.168.40.232
IPs:
  IP: 192.168.40.232
Containers:
  fb-secret-container:
    Container ID:
containerd://d0a922e3a9fa802b4d9c129d36f1ac66ec57300c800638b3a53482967bdb2db5
    Image:            nginx
    Image ID:
docker.io/library/nginx@sha256:1df1a963b5b8a3315c81c9d3c6fbd3324479a16c5e266bc6642c080be636145f
    Port:             80/TCP
    Host Port:        0/TCP
    State:             Running
      Started:         Thu, 11 Jun 2026 16:50:10 +0530
    Ready:             True
    Restart Count:     0
    Environment Variables from:
      fb-secret        Secret Optional: false
    Environment:       <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-rh9rj
```

(ro)

```
Conditions:
```

Type	Status
PodReadyToStartContainers	True
Initialized	True
Ready	True
ContainersReady	True
PodScheduled	True

```
Volumes:
```

```
  kube-api-access-rh9rj:
    Type:              Projected (a volume that contains injected data
from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    Optional:            false
    DownwardAPI:         true
QoS Class:              BestEffort
Node-Selectors:         <none>
Tolerations:            node.kubernetes.io/not-ready:NoExecute op=Exists
for 300s
                      node.kubernetes.io/unreachable:NoExecute
for 300s
```

```
op=Exists for 300s
```

```
Events:
```

Type	Reason	Age	From	Message
----	-----	----	----	-----
Normal	Scheduled	98s	default-scheduler	Successfully assigned default/fb-pod-with-secret to ip-192-168-50-122.ec2.internal
Normal	Pulling	99s	kubelet	spec.containers{fb-secret-container}: Pulling image "nginx"

```
Normal Pulled 96s kubelet spec.containers{fb-secret-
container}: Successfully pulled image "nginx" in 2.839s (2.839s including
waiting). Image size: 63125971 bytes.
Normal Created 96s kubelet spec.containers{fb-secret-
container}: Created container: fb-secret-container
Normal Started 96s kubelet spec.containers{fb-secret-
container}: Started container fb-secret-container
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ ll

total 15

```
-rw-r--r-- 1 Bhavana 197121 691 Jun 9 19:01 1-pod_rr_rl.yaml
-rw-r--r-- 1 Bhavana 197121 481 Jun 9 23:31 2-postgress-pod.yaml
-rw-r--r-- 1 Bhavana 197121 514 Jun 9 23:29 3-pod-env.yaml
-rw-r--r-- 1 Bhavana 197121 295 Jun 11 16:34 4-fb-configmap.yaml
-rw-r--r-- 1 Bhavana 197121 273 Jun 9 23:49 5-fb-pod.yaml
-rw-r--r-- 1 Bhavana 197121 184 Jun 11 16:35 6-secret.yaml
-rw-r--r-- 1 Bhavana 197121 268 Jun 11 16:32 7-fbpod-withsecret.yaml
-rw-r--r-- 1 Bhavana 197121 259 Jun 11 16:45 8-appconfigmap.yaml
-rw-r--r-- 1 Bhavana 197121 594 Jun 11 16:44 9-backendDeploy.yaml
```

#8-appconfigmap.yaml

apiVersion: v1

kind: ConfigMap

metadata:

name: app-config

data:

App_Name: "My Food Truck Application"

App_Version: "1.0.0"

App_Description: "This is a Food Truck application configuration Data"

App_Environment: "Production"

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl apply -f 8-appconfigmap.yaml
configmap/app-config created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get cm

NAME	DATA	AGE
app-config	4	7s
kube-root-ca.crt	1	19m

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl describe cm app-config

Name: app-config

Namespace: default

Labels: <none>

Annotations: <none>

Data

====

App_Description:

This is a Food Truck application configuration Data

App_Environment:

Production

App_Name:

My Food Truck Application

App_Version:

1.0.0

BinaryData

=====

Events: <none>

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get secret

NAME	TYPE	DATA	AGE
fb-secret	Opaque	3	3m24s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl describe secret fb-secret

Name: fb-secret
Namespace: default
Labels: app=fb-app
Annotations: <none>

Type: Opaque

Data

=====

Batch: 4 bytes
Course: 10 bytes
Trainer: 10 bytes

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
fb-pod-with-secret	1/1	Running	0	4m16s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl delete po fb-pod-with-secret

pod "fb-pod-with-secret" deleted from default namespace

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ ll

total 15

-rw-r--r--	1	Bhavana	197121	691	Jun	9	19:01	1-pod_rr_r1.yaml
-rw-r--r--	1	Bhavana	197121	481	Jun	9	23:31	2-postgress-pod.yaml
-rw-r--r--	1	Bhavana	197121	514	Jun	9	23:29	3-pod-env.yaml
-rw-r--r--	1	Bhavana	197121	295	Jun	11	16:34	4-fb-configmap.yaml
-rw-r--r--	1	Bhavana	197121	273	Jun	9	23:49	5-fb-pod.yaml
-rw-r--r--	1	Bhavana	197121	184	Jun	11	16:35	6-secret.yaml
-rw-r--r--	1	Bhavana	197121	268	Jun	11	16:32	7-fbpod-withsecret.yaml
-rw-r--r--	1	Bhavana	197121	259	Jun	11	16:45	8-appconfigmap.yaml
-rw-r--r--	1	Bhavana	197121	594	Jun	11	16:44	9-backenddeploy.yaml

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get cm

NAME	DATA	AGE
app-config	4	11m
kube-root-ca.crt	1	30m

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get secret

NAME	TYPE	DATA	AGE
fb-secret	Opaque	3	14m
fb-secret	Opaque	3	16m

#9-backendDeploy.yaml

```
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-deploy-pod
spec:
  replicas: 3
  selector:
    matchLabels:
      app: backend-app
  template:
    metadata:
      labels:
        app: backend-app
    spec:
      containers:
        - name: backend-deploy-container
          image: nginx
          ports:
            - containerPort: 80

          env:
            - name: BackendApplication
              value: "This is Backend Application"

          envFrom:
            - configMapRef:
                name: app-config

            - secretRef:
                name: fb-secret
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl apply -f 9-backenddeploy.yaml
deployment.apps/backend-deploy-pod configured

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

\$ kubectl get all

NAME	READY	STATUS	RESTARTS	AGE
pod/backend-deploy-pod-76f5ff664c-blmvs	1/1	Running	0	8s
pod/backend-deploy-pod-76f5ff664c-z67xm	1/1	Running	0	7s
pod/backend-deploy-pod-76f5ff664c-zrs7m	1/1	Running	0	3s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	35m

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/backend-deploy-pod	3/3	3	3	6m4s

NAME	DESIRED	CURRENT	READY
AGE			
replicaset.apps/backend-deploy-pod-76f5ff664c	3	3	3
11s			
replicaset.apps/backend-deploy-pod-77f6d4c54d	0	0	0
6m5s			

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june_And_9_June (main)

```
$ kubectl exec -it backend-deploy-pod-76f5ff664c-blmvs -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=backend-deploy-pod-76f5ff664c-blmvs
NGINX_VERSION=1.31.1
NJS_VERSION=0.9.9
NJS_RELEASE=1~trixie
ACME_VERSION=0.4.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
App_Description=This is a Food Truck application configuration Data
App_Environment=Production
App_Name=My Food Truck Application
App_Version=1.0.0
Batch=B17D
Course=AWS DevOps
Trainer=MadhuKiran
BackendApplication=This is Backend Application
KUBERNETES_PORT_443_TCP=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.100.0.1
KUBERNETES_SERVICE_HOST=10.100.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.100.0.1:443
TERM=xterm
HOME=/root
```